

Developing a Stereology-based Approach to Quantify Particles in Ultrathin Sections of the Planarian *Schmidtea Mediterranea* via STEM-EDX

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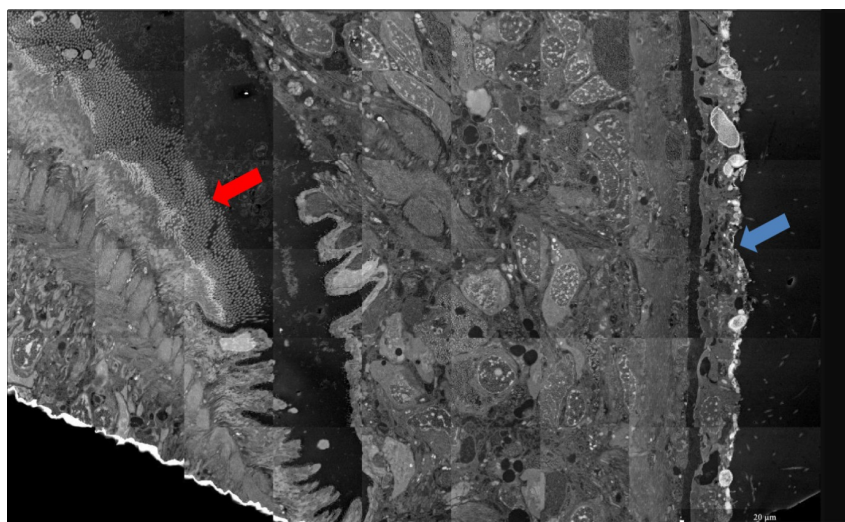
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Abstract

Freshwater planarians are considered interesting alternative models to develop new approach methodologies (NAMs) for (neuro)developmental toxicity assessment of chemicals and particulate materials^{1–3}. They possess an exceptional regenerative capacity due to a high amount of pluripotent stem cells, allowing them to regenerate their entire body, including their central nervous system, within approximately seven days^{4,5}. Further mechanistic insights into (nano)particle toxicity across different stages of (neuro)regeneration can be gained by integrating data from multiple test methods, including scanning transmission electron microscopy (STEM) combined with energy dispersive X-ray spectroscopy (EDX)^{6,7}. This can provide information on (nano)particle uptake, morphology, and agglomeration state, while simultaneously revealing potential ultrastructural changes⁸. In this work, a stereology-based approach is presented for the ultrastructural detection of (nano)particles by STEM-EDX within ultra-thin sections of the planarian *Schmidtea mediterranea*. The procedure is based on the chemical fixation and embedding of specimens in Epon-Spurr resin, which are subsequently sectioned ultra-thin and stained with lead citrate for ultrastructural analysis. Image acquisition, analysis, and workflow efficiency are enhanced by integrating innovative software such as Velox and MAPS (ThermoFisher Scientific, Eindhoven, The Netherlands). These allow automated acquisition of large overview images by stitching smaller high-resolution tiles together, greatly facilitating navigation through complex samples. By combining this with the physical disector principle, an unbiased estimate of the number and volume density of intracellular (nano)particles can be defined. This stereological method will be integrated to objectively quantify (nano)particle uptake and transformations in planarians, evaluating their potential as a NAM for assessing (neuro)developmental toxicity of particulate materials applied in the food chain.



Stitched STEM overview image obtained using Velox and MAPS software of a longitudinal ultra-thin section (60 nm) of the freshwater planarian *Schmidtea mediterranea*, showing the pharynx (red arrow) and epidermis (blue arrow).

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